

Contents

Table of Contents

Contents	1
Table of Contents	1
List of Acronyms	1
List of Figures	1
Introduction	2
An overview of the literature	3
Motivations & Work Objectives	6
Contents of the thesis	7
CHAPTER 1: Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers	9
1.1 Modelling of porous media	9
1.2 Wave propagation in homogeneous layers	13
1.3 Methods for modelling metaporous surfaces and interfaces	18
1.4 Dispersion relations and band structure in periodic lattices	19
Appendices	21
A Benchmark of metaporous configurations	22
B Implementing SAFE to compute the dispersion relation of Lamb modes	22
C Details on the root-finding algorithm	25
D Overview of activities and scientific works during the thesis	27
E Extended abstract in french	28

List of Acronyms

BEM	Boundary Elements Method	PEM	Poroelastic material
EPWE	Extended Plane Wave Expansion	PML	Perfectly Matched Layer
FEM	Finite Elements Method	PWE	Plane Wave Expansion
GEP	Generalized Eigenvalue Problem	SAFE	Semi-analytical Finite Elements
GM	Global Method	SCM	Spectral Collocation Method
HR	Helmoltz Resonator	SRR	Split-Ring Resonator
JCA	Johnson-Champoux-Allard	TMM	Transfer Matrix Method
MST	Multiple Scattering Theory	WFE	Wave Finit Elements

List of Figures

I.1	Periodic cells in 1, 2 and 3D. Image taken from Ref. ²⁷	3
I.2	Example of metaporous configurations	5
I.3	Diagram presenting a general layout for the configurations that will be studied in each chapter	7
1.1	Porous material structure	10
1.2	Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.	13
1.3	Diagram with results of the reflection and transmission coefficients for equivalent fluid and poroelastic layers	16
1.4	Discretization for each method	18
1.5	a) Dispersion relation for the Lamb modes with the root-finding results, alongside the spectral collocation and semi-analytical finite elements solutions. b) Convergence curve of the 2 methods at a given frequency $f = 4$ kHz. The multiple lines depicts the convergence of each wavenumber.	24
1.6	iteration muller	25
1.7	Description of the routine	26

Introduction

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severely affected areas. Sources of noise can be diverse; in urban environments, concerns are mostly on traffic and population noise. From industrial origins, noise can have various consequences, causing not only risks to the workers and the neighbouring populations, but also structural hazard to installations. Generally, motorized transport is a major issue in all environments, including aircraft, public transportation, roads, commercial vessel traffic, etc.

The European Environment Agency estimated in 2021 that 1 in 5 European resident are exposed to unhealthy levels of noise, its main cause being traffic. These noise exposures can have a strong impact on health¹ favoring impaired circadian rhythm, oxidative stress and cardiovascular dysfunction², as well as psychological consequences like anxiety and depression³. Moreover, anthropic noise pollution also affects biodiversity at large, causing stress, behaviour changes, and masking sensory perception⁴. Marine traffic was shown to generate stress on marine mammals and modify their behaviour by provoking area avoidance and vocalization disruption⁵.

Although sound intensity can be measured objectively, the impact of noise on individuals is in part shaped by social and cultural context. Studies show that expected behaviours norms and customs strongly influence reported annoyance⁶, with comparative surveys revealing clear cross-cultural differences. More recent work has linked collectivistic cultural norms to systematically higher everyday noise exposure as an example^{7,8}. Through these examples, we can conclude that noise is a very broad topic that raises complex questions in various scientific disciplines, regarding its impact on human life and biodiversity at large. In this work, we want to focus on noise mitigation using engineered structures in acoustics.

Most noise sources have emerged from the modernisation of societies through various innovations; with them come with the need to mitigate their noise impact up to our current capabilities, which is able to evolve thanks to research on noise barriers and acoustic insulation systems. Various strategies have been applied throughout the ages. Dating back to 4th century BC, acoustic jars were used to improve the room acoustics in Greek and Roman theatres as well as in the Middle Ages in Western Europe in churches⁹. These resonators are thought to be capable of reducing reverberation time in buildings, covering a frequency range matching that of human spoken voice. As another example, a famous roman amphitheater was shown to have a seat row disposition able to improve intelligibility of speeches from the stage¹⁰.

Contemporary to these antique devices, porous and fibrous materials have been traditionally used to absorb sound in the last century¹¹⁻¹⁶. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature¹⁷⁻²¹. They can be found in natural occurrences or engineered and cover very large use cases of acoustic insulation such as industry, transport, aerospace, concert halls and many others.

An overview of the literature

A brief history of Acoustic Metamaterials

In more recent years, the emergence of more complex engineered structures has disrupted the landscape of acoustic absorbers. Metamaterials are composite structures designed to exhibit physical effective properties that go beyond the capability of the sole material that composes the bulk. They can be created by a lot of various means, including engineered microstructure, a gradual evolution of the material properties, or repeating patterns or scatterers implemented in one or several directions in space and embedded in a host material as shown for various dimensions in Fig. I.1. Compared to homogeneous layers, they generally lead to more complex structures, and are able to achieve enhanced effects. Usually allowing for a tunable response, these structures offer great flexibility and open new perspectives in material design. This field has been initiated by Veselago, who has theoretically described the electrodynamics of materials having a negative refraction index (occurring in the case of a doubly-negative permittivity and permeability), now known as the Veselago lens in electromagnetic metamaterials²². These discoveries have subsequently sparked interest in various disciplines, such as photonics, optics²³ and in the case that concerns us, acoustics. Metamaterials can be used to design antennas and absorber for electromagnetic waves and seismic protection devices among others. In acoustic, they allow for various applications such as absorption, filtering, diffusion or cloaking.

Seminal works have been done on the study of the use of lattice arrangements or crystals for wave control applications. Photonic crystals are used to achieve the control of light and electromagnetic waves²⁴. Similarly in acoustics, phononic crystals have been developed for the control of elastic and acoustic waves as the 2D (or 3D) periodic distribution of scatterers in a solid host medium²⁵ as depicted on Fig. I.1. In the case where the host medium is constituted by a fluid, these structures are called sonic crystals²⁶. These structures are often studied through their band structure (also called dispersion diagram/relation when applied to the study of non-periodic geometries), which will be discussed in extensive details in Section [ref this in chap1](#). In the remaining of the literature overview, focus will be centered around metamaterials that involve periodicity in only one direction in most cases. These acoustic metamaterials can be divided into two main categories: the first concerns the so-called *reflection problem* where the case study is that of an incoming wave on some sample or panel which has a rigid-backing at the end. There, the consideration is focused on mitigating the reflected wave and achieving important acoustic absorption. The other case is that of a *transmission problem* where the incoming wave is able to transmit through the sample. In this case and depending on the application, either or both reflection and transmission are sought to be mitigated. In each of these cases respectively, the structures can be called metasurfaces and interfaces.

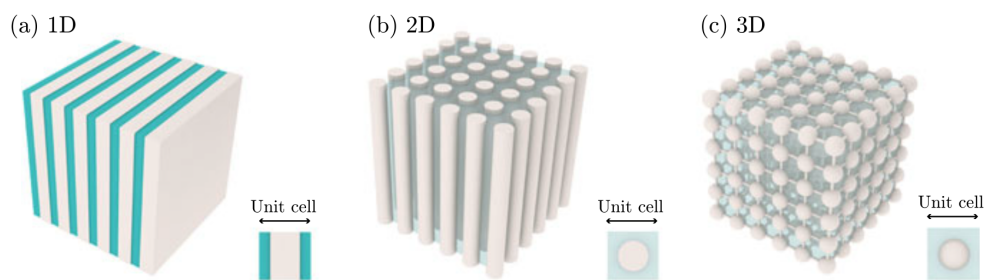


Fig. I.1: Periodic cells in 1, 2 and 3D. Image taken from Ref.²⁷

Structured materials that tackle the reflection problem can involve many various strategies. The combination of a perforated plate and coiled-up channels is able to achieve large broadband absorption underwater²⁸. Micro-perforated plate can also be used to achieve interesting absorption

properties for airborne acoustics²⁹. Materials structured from resonant building blocks with resonance frequencies in the sub-wavelength regime i.e., for frequencies which wavelength is much bigger than the typical size of the material, have also been studied^{30–32}. Large width elastic bandgaps have been achieved using locally resonant cells in the design of a sonic crystal³³. These resonances introduce strong dispersion of the waves and the properties of the system are thus dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions^{34,35}. Balancing the dissipation and leakage of each resonance yields the recently described *critical coupling*. In this case, the reflection drops to zero³⁶. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves^{34,37} and broad range of frequencies^{37–41}. This can be achieved by tuning the structure using the scattering matrix zeros plotted in the complex frequency plane; critical coupling would be achieved when the scattering matrix zero has a vanishing imaginary frequency. Similarly to these broadband absorbers, Yang *et al.* have realized the backing of a porous layer using Fabry-Pérot channels to design an efficient structure⁴².

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been widely developed in the literature, a small subset of these works include Refs. ^{43–46}. These configurations are based on resonant elements and display a zero transmission in the lossless case at their resonance frequency. In reality, losses cannot be avoided and a small amount of the energy is always transmitted. Membrane-type acoustic metamaterials^{47–49}, or acoustic metamaterials made by Helmholtz resonators^{50–52} have been used to develop systems with high values of transmission loss for audible acoustics. Considering the vibroacoustic coupling allows for more precise depiction of real-life applications. Works on periodic soft inclusion in a soft solid matrix was done by Aberkane *et al.* showing improved transmission loss affected by the presence of soft inclusions in the solid layer. In this case, the presence of losses was able to improve the transmission loss at low frequencies. Also accounting for vibroacoustic coupling, attached resonators to walls have been also used to reduce the transmission loss of panels^{53,54}.

For the case of mirror-symmetric structures, it is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, then only 50% of energy can be absorbed as shown in several articles^{55,56}. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances^{47,50,57}, although requiring for a compromise between absorption and thickness. A more complex approach consists in designing degenerate resonators. By using combinations of membrane and cavity resonators^{47,58} and with slits loaded with Helmholtz resonators⁵⁶, quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides⁵⁵. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side^{34,37,49}. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber³⁸.

From porous layers to metaporous surfaces

As mentioned in the beginning of the introduction, porous materials have been traditionally widely used for their acoustic insulation performances. These materials are lossy: an airborne acoustic wave propagating through a porous layer is dissipated as it propagates in the bulk in the form of

viscous friction due to the drag between the air and the pore walls and in the form of heat due to the small ducts and the existence of a thermal boundary layer. These losses mechanisms are strongly dependant on the nature of the material, as they are mainly affect by the geometry of the microstructure. Common models usually account for homogenized effective parameters to characterize these materials as will be discussed in a further section. Since wave velocity in porous media tends to be smaller than that of air, allowing to lower slightly the frequency of the quarter-wavelength resonance, which corresponds due the frequency where such materials usually start to have an efficient acoustic absorption. Section 1.1 is dedicated to presenting the various models used for porous and poroelastic media modelling. Their absorption efficiency depends thus on their inherent losses but also on the impedance mismatch between the layer and the medium from which the impinging acoustic wave is originated (i.e usually air). The acoustic impedance mismatch leads to reflection at the interface, while the energy transmitted through the interface in the porous layer gets attenuated. In real-life applications, these materials are not closed systems and there is a need to account for their coupling irl to surrounding domains into which part of the acoustic energy is susceptible to be radiate back into the half-space.

In order to increase the capabilities of acoustic absorption of porous materials, some works can involve the modification of the core structure of porous and poroelastic materials by tuning of the microstructure, introducing a gradient of the properties, or by using anisotropic materials. Recently, with the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties^{59,60}, or on anisotropic materials^{61–64}. Recently, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption⁶⁵. Other configurations have been studied, among them the periodic partitioning of the metaporous layer⁶⁶ or a labyrinthine channel embedded in a porous host⁶⁷, based on the idea of coiling up the space of the channel as was done for non-porous metamaterial designs. Recently, a hierarchical multilayer metaporous structure was proposed using an optimization procedure, giving rise to gradient and periodic designs⁶⁸.

Some studies involving porous layers embedding periodic arrays of inclusions were done during the last 15 years^{69–72}. These inclusions can have either resonances or not and they allow for an increased dissipation in the viscous regime. These structures are refered to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems^{73,74}. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness⁷⁰. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs⁷⁵. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary^{71,75}. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled⁶³, designed and experimentally characterized showing perfect absorption peaks. More complex spiral-shaped slits have been embedded in a porous layer and have shown to be able to maintain a high absorption for large incidence angles⁷⁶. Xiong *et al.* have studied a metaporous liner with embedded resonator and shown the existence of an exceptional point when the geometry is tuned properly, leading to enhanced absorption⁷⁷.

Fig. I.2: Example of metaporous configurations

Previous metaporous layers studies have considered a host medium modelled as a rigid-frame porous layer; extensions of these structures to consider poroelastic materials have been done⁷⁸. Interesting properties were exhibited for thin elastic shell inclusions, and particularly in the case of a coupling between the trapped and the first volume mode leading to a high wide band absorption. While this previous study implements a semi-analytical model that makes use of Multiple Scattering

Theory (MST), it is worth noting here that different modelings exist as well, based on a Finite Elements Method (FEM) and MST have been validated for the case of an air-saturated metaporoelastic configuration with a solid inclusion and rubber coatings⁷⁹. This thesis is especially focused on the computation of the dispersion relation of such configurations. Therefore, a dedicated literature review dealing with the tools available to achieve such computation will be given in Chapter 1.

Wave propagation modelling and dispersion analysis

The analysis of wave propagation in periodic and complex media often relies heavily on numerical tools. Two main topics are of interest in this thesis: the scattering of an incident wave by a structure as well as its dispersion relation. The first one is simply the acoustic response of a system, where an incident wave propagates towards a specific geometry and is scattered by the latter. The scattering properties are of utmost importance, since tuning those is the main topic in metamaterial and the goal is often to reach exceptional properties for these, such as perfect absorption of the incident wave by the structure as an example. These effects can occur because a specific mode of the structure is excited. These scattering properties depend on the nature of the material(s) used, on the geometry (as we have seen, periodicity can strongly affect the propagation) but also the modal behaviour of the whole system. In order to understand in more details this behaviour, we must turn to dispersion relations.

Dispersion relations are diagrams that represent the wavenumber in a given direction against the frequency, thus informing on the evolution of the wave velocities in the structure and thus the dispersion within. In the context of metamaterials and periodic structures, they are often referred to as band structure. A theoretical definition of a dispersion relation consists in the spectrum of all solutions of a set of equations given a set of parameters. It shows the possible existing states of a system. Generally the form of the dispersion relation, and the various modes that are uncovered can give very in-depth details about the modal behaviour of any structures. In ultrasonics and non-destructive testing, localizing zero-group velocity (ZGV) points in the dispersion relation is very helpful for a number of uses including defects localization and material characterization⁸⁰.

Scarce literature focuses on the computation of the dispersion relations of dissipative layers in general exists: the dispersion of waves in a single poroelastic layer has been studied by Boeckx *et al.*^{81,82} while an unpublished paper from Kiefer *et al.* took interest in Lamb waves in viscoelastic layer coupled to surrounding half-spaces⁸³. There is an interest in analogously studying the coupling between surrounding media and metaporous surfaces, as part of the acoustic energy of the wave propagating in the surface should radiate back into the domain, leading to more complex loss effects on the structure.

These aspects will be discussed in detail in the following chapter, both from the theoretical perspective of porous and poroelastic media (and their coupling to surrounding media) and in terms of the available numerical tools for their computation.

Motivations & Work Objectives

This thesis is dedicated to the numerical modeling of wave propagation in open-system structures comprised of stacks of layers, homogeneous or with embedded scatterers. As it has been seen in the previous section, these materials combine the dissipative characteristics of porous media with the geometric or structural features typical of acoustic metamaterials, thus offering advanced control over their scattering properties. Understanding and predicting the acoustic behavior of such systems is therefore essential for designing efficient structures. In particular, this work focuses in part on the calculation of the dispersion of waves in these structures, and understand how the propagating and evanescent modes that exist in these configurations are shaping the scattering properties. Thence, the

development of efficient numerical strategies is critical to this study.

The tuning of metamaterials could be achieved in some studies by looking at the zeros of the reflection coefficient in the complex frequency plane, thus highlighting the zeros of the scattering matrix. However, the wavenumber might be able to give out additional informations regarding the acoustic behaviour of metaporous layers. Band structure gives useful informations concerning sonic and phononic crystals. However, the dispersion relation of a metaporous layer embedded in an open system has never been addressed; both mechanisms of energy leakage in the surrounding media and dissipations in the porous layer should be investigated. However, this imply a heavy modification of the boundary conditions of the structure. It expected to have significant consequences on the propagation of modes therein. This work will be done on both homogeneous layers as well as periodic structures.

From these observations, we should review existing numerical methods able to compute the acoustic response of metaporous structures and consider ways to obtain a precise and robust method able to predict the scattering properties and acoustic responses of metaporous surface and interfaces. Additionally, we should evaluate methods that have been implemented for the computation of guided waves in layers and stacks of layers coupled to semi-infinite media. Confronting the dispersion relation, with the absorption coefficient should gelp highlighting the modal behaviour responsible for the various absorption peaks obtained by previously studied configurations and also highlight the full modes existing in the layers which might uncover additional phenomena.

Contents of the thesis

This introductory section has set the context as well as an overview of the literature in which this work is situated. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient that the traditional foams used for sound insulation. In particular, metaporous surfaces have shown to be promising options achieving one or multiple unitary absorption peaks at subwavelength frequency. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of these configurations. Thus, in this thesis the main interest and final is to succesfully compute the dispersion relations of guided waves in such structures. The manuscript is split into four different chapters as follows,

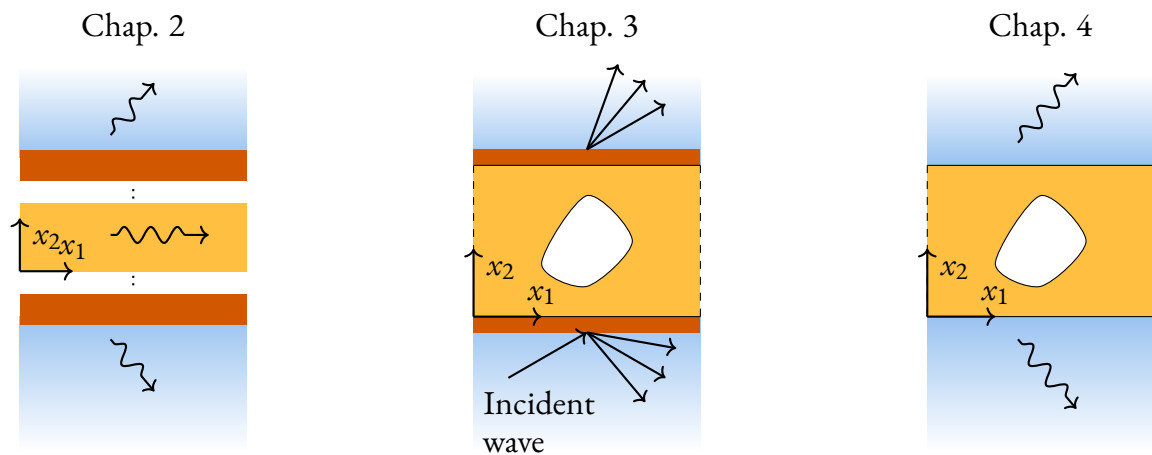


Fig. I.3: Diagram presenting a general layout for the configurations that will be studied in each chapter

- CHAPTER 1 introduces the fundamental concepts necessary for the good understanding of the thesis. This chapter focuses on the scattering of waves on dissipative layers and the guided waves formalism that is used for the calculation of dispersion relations. Additional bibliography is

proposed on these topics as well as a summary of usable numerical methods to compute such problems.

- CHAPTER 2 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A global method is coupled to a FEM. Results focused on the performance of the proposed method, compared to other similar frameworks is proposed. The convergence of this method is shown for various metaporoelastic surfaces, which geometries were studied in previous papers. This chapter was originally an article submitted to *Int. J. Numer. Methods Eng.*^{P1}.
- CHAPTER 3 targets the study of the propagation of guided waves in multilayer arbitrary structures with dissipative layers. A simple numerical method, the Spectral Collocation Method (SCM) is implemented and extra steps allow for the local coupling of surrounding media allows for the computation of the dispersion relation for these configurations. This chapter is constituted from paper^{P2}.
- In CHAPTER 4, the modal behaviour of metaporous structure is studied. The fem method from the previous chapter is adapted to describe this problem. Results are presented for the case of a rigid inclusion, as well as with embedded Split-Ring Resonator (SRR). Contents for this chapter have been submitted as an article.

Finally, a summary of this work is proposed along with some perspectives on this work, including early results generalizing results from chapter four to include poroelastic hosts.

Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers

Contents

1.1	Modelling of porous media	9
1.1.1	Rigid-frame approximation	10
1.1.2	Elastic motion of the skeleton: Biot theory	11
	Formulations for the motion equations	11
	Propagation in unbounded poroelastic media	12
1.2	Wave propagation in homogeneous layers	13
1.2.1	Analytical plane wave modelling	14
1.2.2	Transfer Matrix Method	15
1.2.3	Guided waves for dispersion problems	16
	Computing the modes with semi-analytical tools	17
	Numerics for mode computations	17
	A bestiary of the wave solutions in dissipative and open waveguides	17
1.2.4	Numerics for the computation of dispersion relations	18
1.3	Methods for modelling metaporous surfaces and interfaces	18
1.3.1	A word on periodic structures	18
1.3.2	Scattering of waves by periodic structures: overview of methods	19
1.4	Dispersion relations and band structure in periodic lattices	19
1.4.1	Unpromising adaptation of existing methods	20

This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 | Modelling of porous media

Porous media is a category of materials which encompasses naturally occurring substances such as rocks, soils and biological tissues, and engineered materials like ceramics, metallic foams, rockwool or certain types of polymers. They are considered biphasic, that is constituted of two distinct domains: a pore network Ω_f filled by a saturating fluid embedded in an elastic matrix Ω_s . A diagram of this

microscopic structure is shown in Fig. 1.1. The porosity is defined as the fraction of fluid over the total volume of the material, as $\phi = V(\Omega_f)/V(\Omega)$. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. At microscopic scales, the geometry of the fluid phase thus forms a complex microstructure. Fortunately, at airborne acoustic scales, these materials can often be considered at a meso-/macroscopic level and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that accounts for their complex geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through friction (viscous losses) along the duct/pore network, while thermal exchanges and elastic dissipation take place when the skeleton and the fluid phase are coupled. These losses are thus dependant on the geometry, which parametrized by the porosity, the saturating fluid properties, the pore network size, connectivity and shape, as well as the frame mechanical parameters.

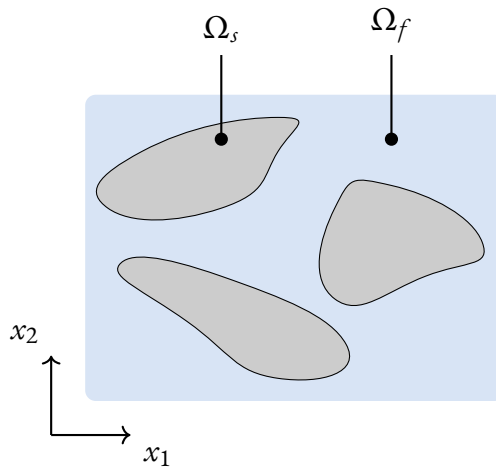


Fig. 1.1: Porous material structure

1.1.1. Rigid-frame approximation

In the so-called rigid-frame approximation, the skeleton of the porous material is considered motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{i\omega t}$ throughout this manuscript (with the exception of Chapter ??). The Helmholtz equation governs the propagation,

$$(\nabla^2 + k_{eq}^2) p(\mathbf{x}) = 0, \quad (1.1)$$

where p is the pressure field and k_{eq} is the wavenumber of the equivalent fluid. In the case where porous or fibrous materials have a porosity close to 1 and the frequency range of interest is limited, the Delany-Bazley model was originally developed in 1970 to propose empirical expressions for the values of the complex wavenumber and characteristic impedances^{11,84} and later improved by Miki⁸⁵. Although limited in accuracy and validity domain, these models have been widely used since they only required the flow resistivity of the porous material. Later on, a more precise semi-phenomenological model additionally accounting for the viscous and inertial dissipative effects was developed from 1987 to 1991. It is the Johnson-Champoux-Allard (JCA) model which models the complex effective density $\rho_{eq}(\omega)$ accounts for the frequency dependant viscous losses and effective bulk modulus of

the material K_{eq} accounting for thermal exchanges in the layer as,

$$\begin{aligned}\rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\ K_{eq} &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}.\end{aligned}\quad (1.2)$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f , kinematic viscosity μ_f , pressure P_0 and Prandtl number $\text{Pr} = c_p \mu_f / k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f \phi / (\rho_f \alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2. Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is can be the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory⁸⁶⁻⁸⁹ is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works¹¹.

Before presenting the full Biot model, some quantities should be provided additionally to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1 - \phi) \rho_s$. The bulk moduli are denoted K_b and K_s , respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\tilde{N} = N(1 + i\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is thus $\rho_2 = \phi \rho_f$ and the inertial interaction density defined by $\rho_{12} = -\phi \rho_f (\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. ??) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation^{86,88,89} depend on the solid displacement vector $\mathbf{u}_s$ and the fluid displacement $\mathbf{u}_f$,

$$\begin{aligned}\nabla \cdot \boldsymbol{\sigma}_s(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{11} \mathbf{u}_s - \omega^2 \tilde{\rho}_{12} \mathbf{u}_f, \\ \nabla \cdot \boldsymbol{\sigma}_f(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{12} \mathbf{u}_s - \omega^2 \tilde{\rho}_{22} \mathbf{u}_f,\end{aligned}\quad (1.3)$$

where $\boldsymbol{\sigma}^s$ is the solid stress tensor, $\boldsymbol{\sigma}^f$ the fluid stress tensor and the apparent densities of the solid-phase is $\tilde{\rho}_{11}$, that of the fluid is $\tilde{\rho}_{22}$ and the coupling apparent density is $\tilde{\rho}_{12}$. They are complex quantities expression that also account for the contribution of viscous forces and are given as,

$$\tilde{\rho}_{22} = \phi^2 \tilde{\rho}_{eq}, \quad \tilde{\rho}_{12} = \rho_2 - \tilde{\rho}_{22}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}.\quad (1.4)$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned}\sigma_{s,ij} &= [(\tilde{P} - 2\tilde{N})\nabla \cdot \mathbf{u}_s + \tilde{Q}\nabla \cdot \mathbf{u}_f] \delta_{ij} + 2\tilde{N}\epsilon_{s,ij}, \\ \sigma_{f,ij} &= (\tilde{Q}\nabla \cdot \mathbf{u}_s + \tilde{R}\nabla \cdot \mathbf{u}_f) \delta_{ij} = -\phi p,\end{aligned}\quad (1.5)$$

where $\nabla \cdot \mathbf{u}_s$ and $\nabla \cdot \mathbf{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and ϵ_s is the solid-phase strain tensor. The Biot coefficients are denoted as $\tilde{P}$ for the solid elastic coefficient, $\tilde{Q}$ for the elastic-fluid coupling coefficient, and $\tilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\tilde{P} = K_b + \frac{4}{3}N + (1 - \phi)^2 \frac{\tilde{K}_{eq}}{\phi}, \quad \tilde{Q} = \tilde{K}_{eq}(1 - \phi), \quad \tilde{R} = \phi \tilde{K}_{eq}. \quad (1.6)$$

The full coefficients are generalized forms required when the limit $K_b/K_s \rightarrow 0$ no longer holds, as is the case for water-saturated PEM.

A different formulation, especially convenient for finite-elements discretization is known as the $(\mathbf{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\hat{A} = \tilde{P} - 2\tilde{N}_s - \frac{\tilde{Q}^2}{\tilde{R}}, \quad \hat{\rho} = \tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}^2}{\tilde{\rho}_{22}}, \quad \beta = \phi \left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}} \right). \quad (1.7)$$

β is another coupling coefficient, also denoted as $\tilde{\gamma}$ in the literature. Using the second row of Eq. (??) allows for the expression of the fluid displacement as, $\mathbf{u}^f = \phi/(\tilde{\rho}_{22}\omega^2)\nabla p - \tilde{\rho}_{12}/\tilde{\rho}_{22}\mathbf{u}^s$ and using that in Eq. (??) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\mathbf{u}_s, p)$ formulation of the Biot theory⁹⁰ is expressed as,

$$\begin{aligned}\nabla \cdot \hat{\boldsymbol{\sigma}}_s + \hat{\rho}\omega^2\mathbf{u}_s + \beta\nabla p &= 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\beta}{\phi^2}\nabla \cdot \mathbf{u}_s + \frac{1}{R}p &= 0,\end{aligned}\quad (1.8)$$

where $\hat{\boldsymbol{\sigma}}_s$ is the *in vacuo* solid stress tensor,

$$\hat{\sigma}_{s,ij} = \sigma_{s,ij} - \phi \frac{\tilde{Q}}{\tilde{R}} p \delta_{ij} = \hat{A} \nabla \cdot \mathbf{u}_s \delta_{ij} + 2N_s \epsilon_s. \quad (1.9)$$

Propagation in unbounded poroelastic media

These two formulations provide equivalent ways of modelling wave propagation in poroelastic media. It appears that PEM supports three waves: a compression φ_p and shear wave ψ_s related to the solid phase as well as an additional compression wave φ_a , related to the fluid phase. In the case of each of these formulations, the fields can be expressed depending on these potentials,

$$\mathbf{u}_s = \nabla \varphi_p + \nabla \varphi_a + \nabla \times \psi_s, \quad \mathbf{u}_f = \mu_p \nabla \varphi_p + \mu_a \nabla \varphi_a + \mu_s \nabla \times \psi_s, \quad (1.10)$$

with μ_p, μ_a, μ_s are coefficients determined by the ratio of solid over fluid displacements ratio. The wavenumbers for each of these waves are, following section 6.5 of Allard and Atalla textbook¹¹ and denoted as $\delta_i, i = 1, 2, 3$ in the text,

$$\begin{aligned}k_{p,a} &= \sqrt{\frac{1}{2} \left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t \right)}, \\ k_s &= \omega \sqrt{\frac{\hat{\rho}}{\tilde{N}}},\end{aligned}\quad (1.11)$$

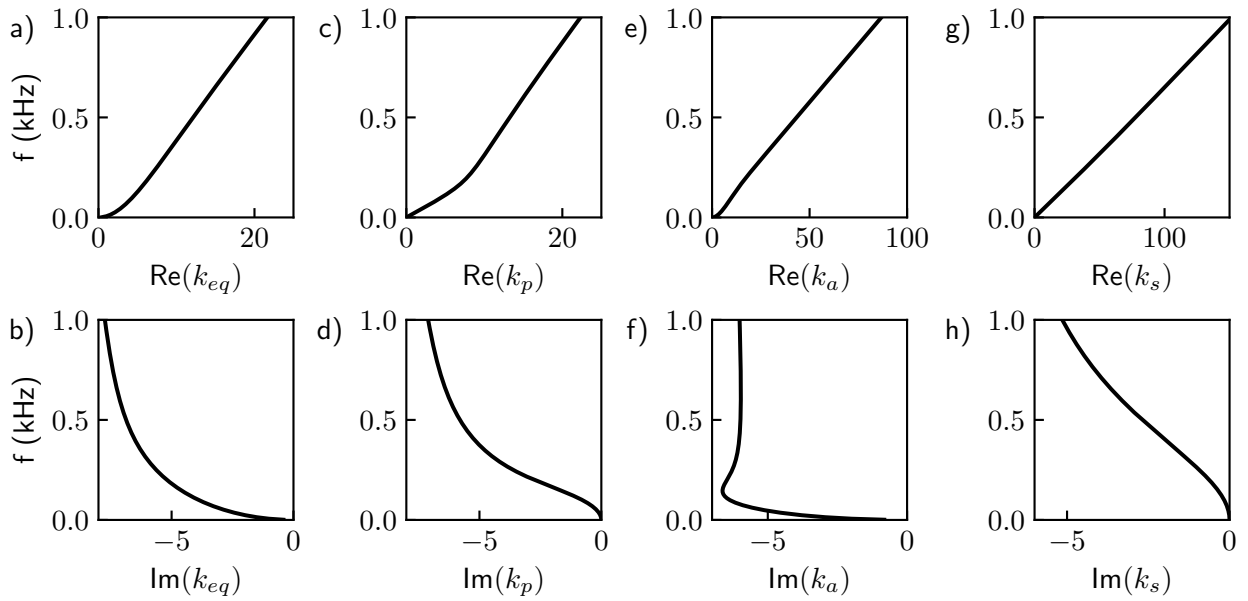


Fig. 1.2: Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.

respectively for the solid compression, fluid compression and shear wave, with,

$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}}, \quad \delta_1 = \omega \sqrt{\frac{\hat{\rho}}{\hat{P}}}, \quad \delta_2 = \omega \sqrt{\frac{\hat{\rho}_s}{\hat{P}}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2 \delta_1^2}. \quad (1.12)$$

The propagation characteristics, i.e. wavenumbers of the waves propagating in porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber for the equivalent fluid as well as the three for the poroelastic layer are complex. Their real part account for the propagation of the phase flow, inversly proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials. It appears that at low frequencies, the attenuation is very low for all waves. The velocity of the compression and shear waves (observed on the figures by the slope of the real parts of the wavenumbers) is much higher than the waves in the fluid for the equivalent fluid and the compression fluid phase velocity, as they propagate in a soft solid where the velocity is usually much higher than in a fluid.

1.2 | Wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, impinging at the surface of a structure and is then scattered (reflected and transmitted) from and through each interface. The existence of an interface leads to the propagation of not only the bulk waves but also of so-called surface waves⁹¹ where the interface itself acts as the support for propagation. Different types of waves propagate along the interface. An example can be surface water waves that follow the interface between air and water; all the energy is localized at the interface. A similar behaviour exist for seismic waves where the interface of a solid

substrate supports the wave propagation, that is Rayleigh (longitudinal motion) and Love waves (transverse motion) which both show an energy decaying with depth. Scholte waves are waves that propagate along the interface between a fluid and a solid medium.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is referred to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often lead to writing complex systems that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various methods available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

1.2.1. Analytical plane wave modelling

Modeling acoustic propagation in homogeneous layers by assuming plane waves is the simplest formalism in acoustics. We assume these plane waves such that they satisfy the motion equations, e.g. the ones introduced in eq. (1.1) or eq. (1.3). Introducing a simple problem, that of the scattering of an incident plane by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\mathbf{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1 , there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements.

Assuming a $e^{i\omega t}$ time-dependence, the incident pressure field originated from the upper domain that satisfies the Helmholtz equation takes the form of,

$$p_{0+} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0}(x_2-h)} + a_- e^{-ik_1 x_1} e^{ik_{2,0}(x_2-h)}. \quad (1.13)$$

The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_2(x_2-h)}, \quad (1.14)$$

and that in the lower air domain is,

$$p_{0-} = c e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \quad (1.15)$$

with wavenumbers such that they hold for the Snell-Descartes law that states $k_1^2 + k_{2,0}^2 = k_0^2$ and $k_1^2 + k_{2,eq}^2 = k_0^2$. The wavenumber should be imposed with $k_1 = k_0 \sin \theta$, $k_2 = -k_0 \cos \theta$, with θ the angle of incidence, that governs the direction of propagation of the incident field. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -i\omega \rho_i u_2, \forall i$.

Next, the interface conditions describe the relationships between properties at the boundary between two media, including the continuity or vanishing of given fields. If a media is terminated by a surface (e.g. a rigid wall), a constraint should be applied on the fields to represent it, denoted as boundary conditions. Assuming a continuity between the two fluids in the present case with interfaces located at $x_2 = 0$ and $x_2 = h$, we obtain the following conditions,

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad \frac{l}{2}(0) = u_2^b(0), \quad (1.16)$$

with superscripts b and t denoting respectively the bottom and top interfaces.

The full analytical description of a system constituted from one or several interfaces can be done by writing a linear system built from the interface conditions, which vector of unknowns are the amplitudes of each of the existing waves in the system, that is $(b_+ \ b_- \ a_+ \ c)^T$. The forcing right-hand side corresponds to the incident wave terms. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients. Imposing the interface conditions leads to writing the following matrix system,

$$\begin{pmatrix} 1 & e^{ik_{2,eq}b} & 0 & -1 \\ \frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}b} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{ik_{2,eq}b} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}b} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ a_+ \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_- \\ \frac{k_{2,eq}}{\rho_{eq}}a_- \end{pmatrix}. \quad (1.17)$$

From this expression, the system can simply be solved to obtain all scattered amplitudes b_+ , b_- , a_+ and c . In the case of a rigidly-backed layer, the interface at $x_2 = b$ is replaced by a rigid termination. For (equivalent) fluids, a hard wall or Neumann-type boundary condition is translated by a vanishing normal particle displacement field along the boundary, that is $u_2(\Gamma) = 0$. The term depending on c thus vanishes, since p_{0-} does not exist; this reduces the size of the system. In this case, the reflection coefficient R corresponds to the amplitude a_+ .

This method is denoted in this work as the Global Method (GM), since all of the available data is used in the formulation of the problem as the unknowns are all wave amplitudes. As described in Section ?? of Chapter ??, a general formalism can be used for this method and any stack of any layer can be described with the GM, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\mathbf{M}(\omega)\mathbf{x} = \mathbf{f}, \quad (1.18)$$

where $\mathbf{M}$ corresponds to the interface conditions involved in the system, $\mathbf{x}$ contain the wave amplitudes and $\mathbf{f}$ is the forcing term, the incident plane wave amplitudes.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in the case of a symmetric configuration (with air coupling at both interfaces) are represented by the diagrams in Figs 1.3a and b and results in Figs 1.3c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^\circ$ angle. The parameters for each material corresponds to the one used in Table ?? from Chapter ??. The system used to obtain the reflection coefficient for the PEM layer is done in a similar fashion as the equivalent fluid and can be read in the appendices of⁷⁸.

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

1.2.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson⁹² and Haskell⁹³ in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics. TMM allows for an easy multiphysics coupling with interface matrices; its core lies in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface, which is possible thanks to

the interface conditions. Then the physical field amplitudes stored in the state vector are propagated from one layer to the next. The transfer matrix T relating the pressure and normal displacement fields from the bottom (located at $x_2 = 0$) to the top interface (located at $x_2 = b$) of a fluid layer is thus,¹²

$$\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} \cos(k_{eq}b) & iZ_{eq} \sin(k_{eq}b) \\ iZ_{eq}^{-1} \sin(k_{eq}b) & \cos(k_{eq}b) \end{pmatrix} \begin{pmatrix} p(b) \\ u_2(b) \end{pmatrix}, \quad (1.19)$$

where k_{eq} is the bulk wavenumber of the wave propagating in the equivalent fluid and Z_{eq} is the medium impedances. The extension to PEM makes of a state vector that includes the fields, $(\hat{\sigma}_{12} \ u_2^s \ u_2^t \ \hat{\sigma}_{22} \ p \ u_1^s)^T$. More details are given in Chapter 11 of Allard and Atalla textbook¹¹.

For stacks of layers, an amount of N layers of the same nature are easy to manipulate since $T_{\text{global}} = \prod_{n=0}^N T_n$ since they have the same size. multiphysics with interface matrix that relate the relation between the various fields at an interface (non square matrices). This method was deep dived in several textbooks among which we propose Refs.^{11,27} for additional details. The TMM also allows for the computation of complex system by an assembly of elements including admittance matrices in order to consider Helmholtz Resonator (HR) embedded in an equivalent fluid layer⁹⁴.

While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes. From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients by simply manipulating the expressions for the fields contained in the state vector with the scattering coefficients to obtain that,

which is only valid for reciprocal and linear assumptions. The absorption coefficient for the same two configurations as the last part is presented in Figs 1.3c and d with the markers. It can be seen that the same results (up to the respective numerical precision of each method) are obtained.

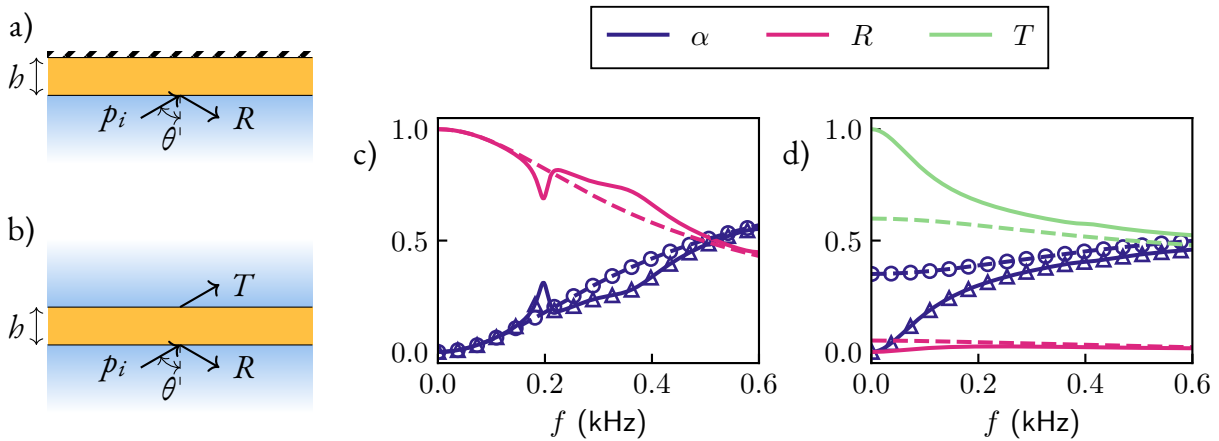


Fig. 1.3: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the c) reflection, d) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid (o) and for the absorption coefficient (Δ)

1.2.3. Guided waves for dispersion problems

From the previously obtained linear systems in Section 1.2.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, representing the guiding of waves in the layer. This set of equations becomes

singular when its determinant vanishes for a given parameter, allowing to identify the modes of the structure. Several formalism can be adopted: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve wavenumbers.

Computing the modes with semi-analytical tools

[finish this](#) for porous/poroelastic materials, root-finding⁸² + groby2008 single homogeneous layers

Once the linear system is obtained, we are faced with either or two of these possibilities. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible.

Using Eq. (1.18), root finding such that $\det(\mathcal{M}(k_1, \omega)) = 0$; the determinant of the matrix from Eq. 1.17 can be expressed as

$$2 \frac{k_{2,eq} k_{2,0}}{\rho_{eq} \rho_0} \cos(k_{2,eq} h) + i \left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2} \right) \sin(k_{2,eq} h) = 0 \quad (1.20)$$

Numerics for mode computations

Regarding the computation of complex modes in homogeneous layer, several numerical methods have been investigated in the last decades. Guided waves represents a large topic and we focus here on the case of simple geometries; similar configurations are considered, an unbounded single (or stack of) layers with two infinite interfaces as presented in the section where we were interested in their acoustic response.

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM), which has been especially used for the computation of modes for guided waves in layers.⁹⁵

In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched, this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include [lister des publis ici](#). Examples are also given in Chapter ?? . A detailed description of the calculation of Lamb modes using a simple SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. ?? . There, its performance is also compared to that of SCM.

[discuss surface and guided waves in a paragraph](#)

A bestiary of the wave solutions in dissipative and open waveguides

In conservative and closed systems, propagating waves are either guided or trapped waves, while in open non-dissipative systems, the acoustic energy is able to radiate outside of the waveguide. Part of propagating waves are thus attenuated and radiate energy to the open domain. They are thus considered quasi-guided since they are propagating waves with some attenuation. Trapped modes can still occur. New family of modes appear as leaky backward and leaky forward waves that exhibit unique properties. Their counterparts are incoming waves.

The addition of dissipation leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation

Fig. 1.4: Discretization for each method

between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished paper is interested in an elastic waveguide⁸³ while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity⁹⁶ [vérifier la citation](#).

There is a discussion to be made on the relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous.

while leading to no assumptions on the radiating field nature contrary to imposing a local coupling, the modeling of the half-space by using some artificial absorbing layer at the end leads to the apparition of modes of the cavities⁹⁷

1.2.4. Numerics for the computation of dispersion relations

- presentation of scm, safe + possible with fem
- cite chap. 2

1.3 | Methods for modelling metaporous surfaces and interfaces

1.3.1. A word on periodic structures

A large literature that shows the enhancement of absorption properties by periodicizing porous layers with scatterers was discussed in the introduction. In this section, a presentation of the fundamental notions surrounding periodic structures is proposed. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of acoustic metamaterials. The periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theorem. This formalism has been pioneered by Felix Bloch and Léon Brillouin among others^{Bloch1929, 98}. Throughout this manuscript, we consider a periodic cell along only the longitudinal direction, that 1D periodicity which we denote d . Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. The layer of infinite extent is thus limited to a single unit-cell. A wavefunction $\psi(\mathbf{x})$ describing the propagation should thus depend on a field with the same spatial periodicity as the lattice. To do so, the wavefunction is projected onto an infinite modal basis, that is Bloch modes,

$$\psi_q(\mathbf{x}) = e^{ikx_1}. \quad (1.21)$$

The wavefunction at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector, that is $\psi(\mathbf{x} + nd) = \psi(\mathbf{x})e^{ikd}$.

The possibility of reduction of the spatial considerations of the problem also leads to a reduction of the reciprocal wavenumber space. In periodic structure, we often restrict this reciprocal space to the first Brillouin zone, that is to wavenumbers restricted to $k_1 \in [0, \pi/d]$ when looking at the band structure. The consequences of periodic media are directly observable in the band structure: bands are usually folded when the wavenumber reaches this limit of the reciprocal space. This folding occurs at the Bragg frequency corresponds to a relation between the wavelength λ of the excitation

signal and the periodicity of the crystal lattices, $d \sin(\theta) = n\lambda$, with θ the angle of incidence. This condition corresponds to destructive interferences: this phenomenon is a cause of the appearance of the famous bandgaps; in these bands of frequency, waves are unable to propagate. **more on bg and stuff**^{stuff here}.

1.3.2. Scattering of waves by periodic structures: overview of methods

Analytical methods for the evaluation of the wave propagation in periodic structures often require tedious derivations, as the frequency regimes of interest is usually high enough to require the description of the multiple scattering happening in the structure. The scatterer of the unit cell onto which the incident wave impinges acts as a secondary source, which scattered field is successively scattered by each neighboring obstacles and so on. Multiple Scattering Theory is a theoretical framework that has been developed to account for these phenomena and was used to compute the acoustic response of metaporous structures with rigid inclusions⁷⁰ and metaporoelastic surfaces with solid and shell inclusions⁷⁸. Both configurations have been tuned to show a perfect absorption peak in the frequency spectrum.

The remaining papers discussing how incident pressure fields are affected and scattered by metaporous structures use FEM, for the most part through the interface of Comsol Multiphysics. This very versatile and powerful method has been used for decades in all fields of physics and thus won't be detailed in this thesis. It is recommended to the reader as an introduction on the basis of FEM, the textbooks from Atalla and Sgard⁹⁹ related to acoustics and vibration problems and that of Dhatt *et al.* for higher-order finite elements problems¹⁰⁰. This method allows for the discretization of the geometry via a mesh in order to solve the appropriate motion equations. This has allowed to use rather complex geometries, e.g spirals¹⁰¹ and labyrinthine paths⁶⁷ as scatterers embedded in porous layers as discussed in the state of the art. Additionally, Gaborit *et al.*⁷⁹ have developed a method in order to couple FEM with Bloch modes expansion of analytical fields described by TMM. They have used it to compute the scattering coefficients of structure that include a periodic cell sandwiched by homogeneous layers with an embedded inclusion.

1.4 Dispersion relations and band structure in periodic lattices

One of the most popular methods for the computation of band structures in periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a Fourier transform of the motion equations, which allows an expression of the fields expanded onto Bloch modes, truncated for numerical applications. This method, while easy to implement presents important limitations due to the convergence of Fourier series and the matching of constituent materials for the lattice: materials should preferably be of the same nature for the calculation to go well²⁷. Generalized as the Extended Plane Wave Expansion (EPWE), the method also allows to obtain the evanescent Bloch modes in Ref.¹⁰², allowing for complex dispersion diagram investigation and to account for evanescent modes in finite 2D periodic lattices.

The complexity of the system prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

1.4.1. Unpromising adaptation of existing methods

The complexity of the system of interest in this thesis as presented in Section?? of the Introduction prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods: as the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells⁷⁰ as well as solid inclusions in a poroelastic unit cell⁷⁸. However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighboring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k_1^X l d) \left((-1)^q e^{i k_1^X l d} + e^{-i k_1^X l d} \right), \quad (1.22)$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

The WFE as well as FEM has also been widely applied to these geometries. Their modularity and versatility lead them to be good candidates for the modelling of our structure.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for "Eigenfrequency" studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the "Weak Form PDE" interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

trucs à rajouter jsp où

summary of all thesis of aberkane (soft solids), magliacano (disp w fem of closed system with porous layers), + gaborit et voir pour les autres
à distribuer au cours du chapitre

Appendices

Contents

A	Benchmark of metaporous configurations	22
B	Implementing SAFE to compute the dispersion relation of Lamb modes	22
	B.1 Formalism	22
	B.2 Results and convergence comparison	23
C	Details on the root-finding algorithm	25
	C.1 The Müller method	25
	C.2 A complete routine for the automated solving of dispersion relations . . .	25
D	Overview of activities and scientific works during the thesis	27
	D.1 Publications	27
	D.2 Conferences	27
	D.3 Workshops and training schools	27
	D.4 Teaching activities	27
E	Extended abstract in french	28

A | Benchmark of metaporous configurations

TODO: 7 ou 8 configurations du benchmark avec coefficient d'absorption COMSOL / py-PLANES (+ MST quand possible ?) avec des schémas

B | Implementing SAFE to compute the dispersion relation of Lamb modes

As discussed in Section. ??, the SAFE method is an alternative to SCM for the numerical computation of dispersion relation of guided waves in layers. They rely on a similar formalism, that is a 1D discretization across the thickness of each layer and an assumption of guided harmonic waves along the longitudinal direction. An analogous method to that presented in Chapter ?? could be developed using SAFE, the key difference being the location of nodes in the layer and the interpolation function used for discretization. There is a large bibliography dealing with these topics and applying this method to similar problem to ours.

The general theory of SAFE is explained in great details in Chapter 9 of Rose's textbook¹⁰³ hence The present method is inspired from. Examples of SAFE application to dispersion curve calculation are illustrated by Predoi *et al.*¹⁰⁴ where a numerical model was implemented to investigate periodic structure of infinite width with the addition of periodic boundary conditions allowing for the computation of guided wave modes. A similar approach was generalized for systems with arbitrary cross-sections and a coupling with infinite solid media by using the so-called absorbing region¹⁰⁵. Other description of the surrounding medium where leakage happens have also been investigated by using a PML¹⁰⁶. This has been done by Mazzotti *et al.* using BEM^{107,108}. A local coupling of the surrounding space was done in Ref.¹⁰⁹ by linearizing a non-linear GEP, a similar approach to that of Refs.^{P2,110}.

In this appendix, we propose to quickly re-derive a SAFE method inspired from these previous papers to benchmark the two numerical methods regarding their respective performances. We will use high-order Lobatto function as interpolating function of the finite-elements core of the method. The system studied here is the simple of Lamb modes: an elastic layer of infinite extent in the x_1 direction has two free edges at $x_2 = 0$ and $x_2 = b$, where a no-traction boundary condition is applied.

B.1. Formalism

Let us start with the general variational form for the elasticity equation

$$-\rho\omega^2 \int_{\Omega} u \delta u \, d\Omega + \int_{\Omega} \sigma(u) : \epsilon(\delta u) \, d\Omega = \int_{\partial\Omega} \sigma(\delta u) \cdot \mathbf{n} \, d\Gamma, \quad \forall \delta u, \quad (\text{B.1})$$

where the displacement field is

$$\mathbf{u}(\mathbf{x}) = \mathbf{u}(x_2)e^{ik_1x_1}e^{-i\omega t}, \quad (\text{B.2})$$

and the double-dot product is $\sigma(u) : \epsilon(\delta u^*) = \sigma_{ij}(u)\epsilon_{ij}(\delta u^*)$. σ designates the stress tensor while ϵ is the strain tensor. Developing this part of the volume term gives the expression,

$$\begin{aligned} \sigma_{11}\epsilon_{11} &= k_1^2(\lambda + 2\mu)u_1\delta u_1 + -ik_1\lambda\nabla u_2\delta u_1 \\ \sigma_{12}\epsilon_{12} &= \frac{\mu}{2} \left(-ik_1\nabla u_1\delta u_2 + \nabla u_1\delta\nabla u_1 + k_1^2u_2\delta u_2 + ik_1u_2\nabla\delta u_1 \right) \\ \sigma_{22}\epsilon_{22} &= ik_1\lambda u_1\nabla\delta u_2 + (\lambda + 2\mu)\nabla u_2\nabla\delta u_2. \end{aligned} \quad (\text{B.3})$$

Putting these terms back into Eq. B.1 yields,

$$\begin{aligned} \int_{\Omega} -\rho\omega^2 (u_1\delta u_1 + u_2\delta u_2) + k_1^2(\lambda + 2\mu)u_1\delta u_1 - ik_1\lambda\nabla u_2\delta u_1 - ik_1\mu\nabla u_1\delta u_2 + \mu\nabla u_1\delta\nabla u_1 \\ + \mu k_1^2 u_2\delta u_2 + ik_1\mu u_2\nabla\delta u_1 + ik_1\lambda u_1\nabla\delta u_2 + (\lambda + 2\mu)\nabla u_2\nabla\delta u_2 d\Omega = 0 \quad \forall \delta u \end{aligned} \quad (\text{B.4})$$

where ∇u_i denotes differentiation of the i -th component of the displacement along x_2 . Note that the surface term initially expressed vanishes from the no-traction boundary condition.

The displacement fields can be discretized as

$$\mathbf{u}(\mathbf{x}) = N(\xi)\mathbf{U}e^{ik_1x_1}e^{-i\omega t}, \quad (\text{B.5})$$

with N containing the Lobatto shape functions, and $\mathbf{U}$ is the vector of displacement amplitudes. Following the same discretization, the test function thus becomes $\forall \delta \mathbf{U}$. By rearranging the terms of Eq. B.4 and writing the weak form for one element,

$$\begin{aligned} \int_{\Omega_e} k_1^2 \left(\mu \delta \mathbf{U}_2^T \mathbf{M}^{(e)} \mathbf{U}_2 + (\lambda + 2\mu) \delta \mathbf{U}_1^T \mathbf{M}^{(e)} \mathbf{U}_1 \right) + ik_1 \left(\mu \delta \mathbf{U}_2^T \mathbf{C}_1^{(e)} \mathbf{U} - \mu \delta \mathbf{U}_1^T \mathbf{C}_2^{(e)} \mathbf{U}_2 \right. \\ \left. + \lambda \delta \mathbf{U}_1^T \mathbf{C}_1^{(e)} \mathbf{U}_2 - \lambda \delta \mathbf{U}_2^T \mathbf{C}_2^{(e)} \mathbf{U}_1 \right) + \mu \delta \mathbf{U}_1^T \mathbf{K}^{(e)} \mathbf{U}_1 + (\lambda + 2\mu) \delta \mathbf{U}_2^T \mathbf{K}^{(e)} \mathbf{U}_2 \\ - \rho\omega^2 \delta \mathbf{U}_1^T \mathbf{M}^{(e)} \mathbf{U}_1 - \rho\omega^2 \delta \mathbf{U}_2^T \mathbf{M}^{(e)} \mathbf{U}_2 d\Omega_e = 0 \end{aligned} \quad (\text{B.6})$$

where the elementary matrices are,

$$\begin{aligned} \mathbf{M}^{(e)} &= \frac{b}{2} \int_{-1}^1 N^T N d\xi, \quad \mathbf{K}^{(e)} = \frac{2}{b} \int_{-1}^1 \nabla N^T \nabla N d\xi, \\ \mathbf{C}_1^{(e)} &= \int_{-1}^1 N^T \nabla N d\xi, \quad \mathbf{C}_2^{(e)} = \int_{-1}^1 \nabla^T N^T N \mathbf{U} d\xi, \end{aligned} \quad (\text{B.7})$$

In Eq. B.6, the dependance on k_1 was highlighted and terms were sorted depending on their order. The elementary matrices should be assembled into the global matrices and then, following an analogous procedure to Chapter. ??, the general form for the problem becomes,

$$\left(k_1^2 \mathbf{A}_2 + ik_1 \mathbf{A}_1 + \mathbf{A}_0 - \omega^2 \mathbf{M} \right) \mathbf{U} = 0. \quad (\text{B.8})$$

It can easily be rewritten as a generalized eigenvalue problem,

$$\left[\begin{pmatrix} -\mathbf{A}_1 & \mathbf{A}_0 - \omega^2 \mathbf{M} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} - k_1 \begin{pmatrix} \mathbf{A}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \right] \begin{pmatrix} k_1 \mathbf{U} \\ \mathbf{U} \end{pmatrix} = 0, \quad (\text{B.9})$$

which solving is thus quite straightforward.

B.2. Results and convergence comparison

Analytical results are obtained by applying the root-finding approach using the Müller method presented in the paper. Solutions are sought from the classical Lamb modes equations, derived in many books such as Royer. Figure 1.5a) shows the results of the three methods for a given frequency range. They all match well. In order to compare the two numerical methods in play, a brief convergence study has been done using the analytical results from the root-finding method as a reference.

The convergence curve of the results given in Fig. 1.5b) is unsurprisingly in agreement with the literature. It is calculated at a frequency $f = 4$ kHz: the solutions along the dashed line in Fig 1.5a)

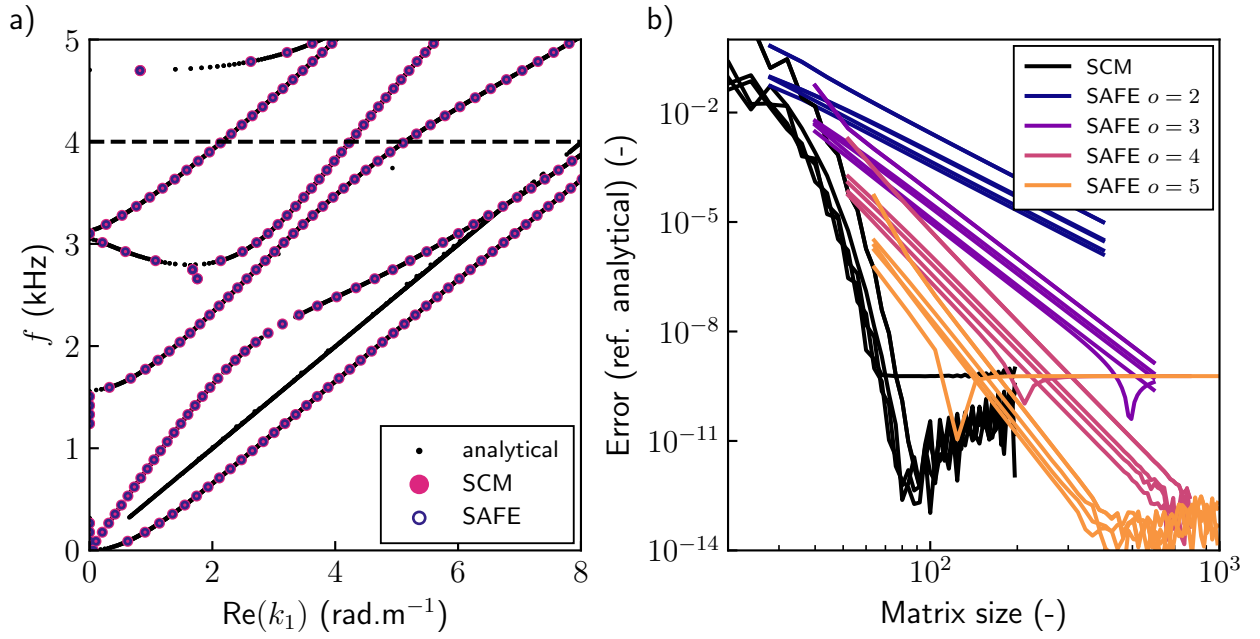


Fig. 1.5: a) Dispersion relation for the Lamb modes with the root-finding results, alongside the spectral collocation and semi-analytical finite elements solutions. b) Convergence curve of the 2 methods at a given frequency $f = 4$ kHz. The multiple lines depicts the convergence of each wavenumber.

are selected. The parameter n used for the discretization is shared between the two methods and corresponds to the number of elements taken for the discretization in each method. SAFE matrices have a size $2(2n + 1)$, while SCM matrices have a size $2n$. The discretization is swept from a number of nodes $n = 3$ to $n = 50$.

Matrix size for scm $4n$ while safe is $4(o \cdot n + 1)$ with o the order of Lobatto shape functions

The SCM convergence shows a steep slope and reaches a plateau around $n = 20$, while the SAFE curve follows the expected slope of a finite elements method that use second-order shape functions. When comparing at a given discretization order, the error from the SAFE is much lower than that of SCM by several orders of magnitude. In order to see comparable convergence, it would require to use higher-order shape functions or use spectral elements instead.

C | Details on the root-finding algorithm

C.1. The Müller method

equations, iterative scheme etc.

¹¹¹

$$\begin{aligned}\Sigma_x &= \prod_{0 \leq i < j \leq 2} (x_j - x_i) = (x_0 - x_2)(x_1 - x_2)(x_0 - x_1) \\ a &= \frac{(x_1 - x_2)(f_0 - f_2) - (x_0 - x_2)(f_1 - f_2)}{\Sigma_x} \\ b &= \frac{(x_0 - x_2)^2(f_1 - f_2) - (x_1 - x_2)^2(f_0 - f_2)}{\Sigma_x} \\ c &= f_2\end{aligned}\tag{C.1}$$

$$f(x_3) = ax_3^2 + bx_3 + c\tag{C.2}$$

$$x_3 = x_2 - \frac{2c}{b + \text{sign}(b)\sqrt{b^2 - 4ac}}\tag{C.3}$$

$$\varepsilon_a = \left| \frac{x_3 - x_2}{x_3} \right|\tag{C.4}$$

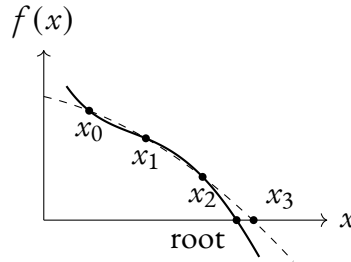


Fig. 1.6: iteration muller

C.2. A complete routine for the automated solving of dispersion relations

In order to find solutions for modes in the form of complex wavenumbers k_1 , we use the previously derived system from Eq. (??), that can also be expressed as $\mathbf{M}(k_1, \omega)\mathbf{S} = 0$. The modes are found by finding the discrete set of k_1 that satisfy the singularity of $\mathbf{M}$ using Eq. ??

In practice, these solutions are found using a root-finding procedure, our own implementation of the Müller method¹¹². It is based on a parabolic interpolation to iteratively finds complex wavenumber solutions k_1 given a specific frequency ω from three initial guesses. Therefore, care has to be put into the initialization of the method. Since the system $\mathbf{M}$ is constructed from a finite elements scheme, computations will be all the more costly that a large number of iterations need to run before convergence is reached. The convergence criterion checks that threshold of the variation of the solution is below ε . In this paper, this criterion is taken with a magnitude around $\varepsilon = 10^{-13}$. Thus, the initialization of the method is done by evaluating the local minimas of a coarse mapping of the function in the complex plane. The latter is only computed every few frequency steps to save

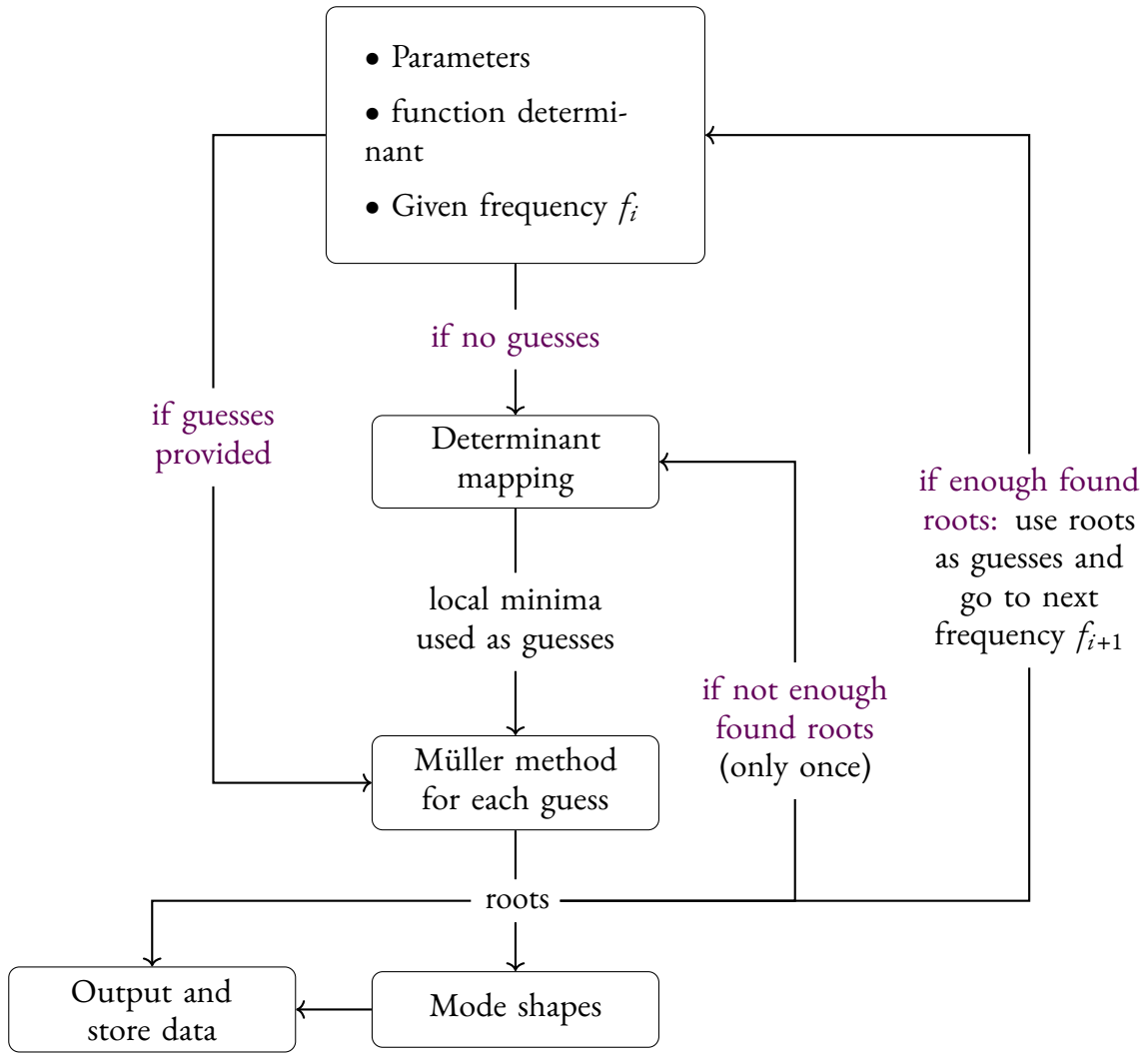


Fig. 1.7: Description of the routine

computation time. Additionally, the matrix D_{ii} from Eq. (??) does not depend on k_1 and can thus be only evaluated once per frequency step, thus saving valuable computing resources.

Once the set of solutions has been obtained, each of the associated mode shapes can also be retrieved. Given a specific root $k_{1,j}$ that satisfies $\det(\mathbf{M}) = 0$, there is $\mathbf{M}(k_{1,j})\mathbf{S} = 0$, where the vector $\mathbf{S} = \ker(\mathbf{M}(k_{1,j}))$ is the nullspace of the matrix. In practice, the nullspace is computed using singular value decomposition (SVD), $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\dagger$. The last right singular vectors, that is the last row of $\mathbf{V}^\dagger$ provide an orthonormal basis for the null space of $\mathbf{M}$, associated to the singular value $\sigma = \min(\mathbf{\Sigma})$. All state vectors among which S^t and S^b are obtained by selecting the appropriate rows of $\mathbf{V}^*$. By doing the whole procedure backwards, using Eq. (??) and then Eq. (??), we can retrieve the full degrees of freedom vector $\mathbf{p}$ (and normalized as $(p^i(x_1, x_2)/p^i(0, h))e^{ik_1x_1}$).

D | Overview of activities and scientific works during the thesis

D.1. Publications

- P1. M. Maréchal, O. Dazel, V. Romero-García, and J.-P. Groby. “Numerical Models for the Scattering Properties of Multilayered Structures with Periodic Cells”. *submitted to Int. J. Numer. Meth. Eng.* (2025).
- P2. M. Maréchal, A. Geslain, J.-P. Groby, V. Romero-García, and O. Dazel. “A General Spectral Collocation Method for Computing the Dispersion Relations of Guided Acoustic Waves in Multilayer Dissipative Structures”. *J. App. Phys.* 137.10 (2025), 104902.

D.2. Conferences

(ajouter titres des présentations)

- SAM 2022 Nimes, France, October 2022 (attendee)
- JJCAB Toulouse 2023, France, July 2023 (poster)
- SAPEM Sorrento, Italy, October 2023 (presenter)
- Séminaire de 2e année Centrale Nantes, France, April 2024 (presenter)
- SAM Aegina, Greece, May 2024 (presenter)
- INTERNOISE Nantes, France, August 2024 (attendee and volunteer)
- CFA 2025 Paris, France (presenter)

D.3. Workshops and training schools

- training school Acoustic Metamaterials Valencia Spain, UPV, November 2023
- lamb waves workshop

D.4. Teaching activities

E | Extended abstract in french

(en 10 à 15 k caractères)

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3. O. Hahad et al. "Noise and Mental Health: Evidence, Mechanisms, and Consequences". *J Expo Sci Environ Epidemiol* 35.1 (2025), 16–23.
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